Jeff Helzner

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Summary

Hands-on decision scientist who builds AI, Bayesian, and machine learning systems that make expert judgment better in high-stakes decisions with incomplete data. Over twelve years in insurance I have designed and shipped production models, data pipelines, monitoring, and LLM tools for underwriting and claims, building the capability from scratch at four organizations. I worked closely with Daniel Kahneman on reducing noise in expert judgment, and before industry I was a professor at Columbia University teaching and researching decision theory, probability, and logic (Ph.D., Carnegie Mellon). A long-standing focus of my work, from graduate school to the present, is how to measure whether decision makers, human or AI, are actually deciding well.

I work end to end, from data ingestion and modeling to the tools and dashboards experts actually use, with a focus on principled uncertainty quantification and on systems that combine model output with human expertise. Fully remote since 2020.

Experience

Director of Data Strategy and Analytics

Berkley Enterprise Risk Solutions (a W. R. Berkley company) *March 2024 – Present*

Hands-on lead for the data systems, probabilistic models, and AI tools behind underwriting and claims decisions at a California-based startup writing large workers' compensation policies. - Designed and deployed Bayesian models that combine class-rate priors with submission-level loss and payroll data to produce principled loss picks; five Bayesian/ML models deployed to production in total. - Built an LLM-powered application that drafts claim status updates for adjusters; adopted broadly across the claims team, with a marked reduction in preparation time. - Built automated monitoring (roughly ten weekly checks across the book of business) that flags anomalous shifts in claim-type frequency or severity for proactive portfolio management. - Built pipelines for ingestion and transformation of underwriting and exposure data supporting submission intake and ongoing claims monitoring. - Built ML pipelines to standardize and impute missing fields in OCR-extracted loss records. - Automated generation of monthly client-facing service booklets, substantially reducing preparation time. - Defined decision-aligned metrics and dashboards with leadership for underwriting and portfolio reviews.

Decision Scientist

Joyn Insurance *October 2020 – January 2024*

At an early-stage insurtech, helped build a data-driven underwriting platform from the ground up, integrating model output with underwriter judgment. - Built ML and Bayesian models to prioritize incoming submissions. - Designed and maintained a software package for managing third-party data used in underwriting decisions. - Built tools and dashboards that surfaced model outputs to underwriters in intuitive, actionable formats. - Partnered with product and engineering to ensure data integrity and model interpretability. - Platform adopted by the underwriting team as its primary submission-triage tool through production rollout.

Decision Scientist, Blackboard Insurance

AIG *December 2018 – August 2020*

Built methods for extracting behavioral insight from operational data at AIG's insurtech subsidiary. - Built behavioral analytics models of underwriter and broker behavior. - Contributed to R&D projects using NLP and OCR to extract structured data from unstructured documents. - Partnered with data engineering on pipelines supporting analytics requirements. - Managed an MS-level data scientist supporting analytics and productionization.

Head of Behavioral Science Team

AIG *May 2014 – September 2018*

Founded and led one of the insurance industry's first behavioral science teams (5 people, including one PhD and two MS-level statisticians), focused on improving the quality of expert judgment in underwriting and claims. - Worked closely with Daniel Kahneman on the design of noise-reduction studies for expert judgment in claims and underwriting. - Ran experimental studies that demonstrated reduced variance in adjuster case-reserve estimates. - Drove process improvements to reduce noise in human judgment and improve underwriting consistency. - Led projects on behavioral analytics and choice architecture with actuarial, claims, underwriting, and shared services. - Presented findings to the CEO and Chief Underwriting Officer; represented the science team at multiple board meetings.

Associate Professor of Philosophy

Columbia University *July 2011 – June 2014*

Assistant Professor of Philosophy

Columbia University *August 2003 – June 2011*

Taught undergraduate and graduate courses on logic, probability, decision theory, and the philosophy of science. Published on decision-making under uncertainty and mathematical logic in *Theory and Decision*, *Synthese*, the *Journal of Behavioral Decision Making*, and the *Archive for Mathematical Logic*. Mentored students who have gone on to careers in data science, economics, software engineering, and academia.

Advisory

- **Decision Science Advisor, Rational AI.** An AI-native firm helping private equity investors with deal sourcing, valuation, and portfolio value creation.
- **Member, BeSci.io Collective.** A collective of senior applied behavioral scientists that helps organizations adopt and scale behavioral science methods.

Research and Open Source

- **Sensitivity to Subjective Expected Utility Maximization: A Methodological Study, with an Illustrative Application to LLM Decision-Making** (arXiv:2607.11920, 2026). A Bayesian framework for measuring how closely a decision maker's choices, human or AI, track expected utility, with adequacy checks for when the measure can be trusted. Open-source Stan/Python workflow: github.com/jeffhelzner/seu-sensitivity.
- Interviewed on the *An N of 1* applied behavioral science podcast (Episode 48, 2026).

Education

Ph.D., Logic, Computation, and Methodology · Carnegie Mellon University

M.S., Logic and Computation · Carnegie Mellon University

M.A., Mathematics · University at Buffalo

B.A., Mathematics · University of Massachusetts

Skills - **Modeling:** Bayesian inference (Stan, PyMC), machine learning (scikit-learn), uncertainty quantification, experimental design - **Applied AI:** LLM application development, prompt engineering, AI-assisted development (GitHub Copilot) - **Data & tooling:** Python, SQL, R, data pipelines, Git, Power BI - **Domain:** Insurance underwriting and claims (workers' compensation, commercial lines), judgment and decision-making, noise reduction in expert judgment - **Interests:** Jazz, ice hockey, football